REG.NO.:

VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION SYSTEMS
CONTINUOUS ASSESSMENT TEST - I
WINTER SEMESTER 2024-2025

SLOT: F1+TF1

Programme Name & Branch	**: Master of Computer Applications**
Course Code and Course Name	**: PMCA606L – Internet of Things**
Faculty Name(s)	**: Balaji. E**
Class Number(s)	**: VL2024250503397**
Date of Examination	**: 01.02.2025**
Exam Duration	**: 90 minutes** **Maximum Marks: 50**

General instruction(s):

- Answer All Questions
- M - Max mark; CO – Course Outcome; BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)

  CO 1- Describe various layers of IoT protocol stack and describe protocol functionalities

  CO 2- Evaluate efficiency trade-offs among alternative communication models for an Efficient IoT application design

Q. No	Question	M	CO	BL
1.	Discuss the IBM's IoT conceptual framework leverage architecture and design principles to enhance scalability, security, and interoperability in enterprise IoT solutions and how do Oracle and IBM differ in their approach to IoT implementation, and what advantages does each framework offer for businesses looking to leverage IoT for data management and analytics?	10	CO1	BL2
2.	Elaborate how an M2M architecture differs from a traditional IoT architecture with neat diagram and discuss the M2M device communication domain entities, components of an M2M Core Network.	10	CO1	BL2
3.	Explain the different levels of IoT with a clear illustration and analyze how IoT Level 4 (Interoperability and System Integration) is applied in smart cities for optimizing traffic management, reducing congestion, and integrating real-time data from multiple sources to improve urban mobility.	10	CO1	BL4
4.	Elucidate how does AMQP facilitate secure, efficient, and scalable communication between smart home devices like security cameras, smart locks, and environmental sensors with data exchange diagram	10	CO2	BL3

	and discuss what advantages does it offer over other IoT messaging protocols for managing a high volume of device-to-cloud interactions?			
5.	A city deploys a CoAP-based smart street lighting system where lights respond to motion sensors. For this scenario, discuss the process steps involved in communication between devices and specify the significance and key roles of the following layers in ensuring efficient delivery in constrained networks: (i) Application Layer, (ii) Messaging Layer, and (iii) UDP Layer.	10	CO2	BL4

****